The Fire Salamander's Secret Glow

From: ScienceDaily Top Environment Mined: 2026-10-29 3 min read

Fire salamanders are easy to spot in Europe. They are black and yellow, and they give off poison to keep hungry animals away. They live in wet forests, near streams, and in mountains. Losing their homes puts them in danger. Now scientists have found a new surprise. Under an **ultraviolet** light, fire salamanders glow! The glow is strongest on the yellow underside of their bodies and along their sides. It comes from **glands** in their skin, and even the liquid from the **glands** can keep glowing for more than 24 hours after it leaves the body. The glow is green and **cyan**. Parts of the skin take in **ultraviolet** light, which people cannot see, and send it back out as light we can see. Fireflies, by contrast, make their own light. Scientists do not yet know why salamanders glow. Maybe the glow helps them find each other or pick a **mate**. Maybe it makes their warning colors stand out more to **predators**. Bernat Burriel, the study's first author, says it is fascinating that such a well-studied animal still holds secrets.

Word Treasure

ultraviolet -- Ultraviolet is a kind of light that people cannot see with their eyes.
glands -- Glands are small parts in skin or a body that make a liquid or substance.
cyan -- Cyan is a bright blue-green color.
predators -- Predators are animals that hunt and eat other animals.
mate -- A mate is an animal partner chosen for having babies.

Background you need

A fire salamander is a small animal with smooth, damp skin that lives in wet places like forests and near streams.

Its skin has glands that can make substances to protect it from hungry animals.

Some living things give off light, and scientists study glowing skin to learn how animals warn others or find each other.

Break it down

WHO: Fire salamanders and the scientists studying them, including Bernat Burriel.
WHAT: The discovery that fire salamanders glow under an ultraviolet light.
WHERE: In Europe, in wet forests, near streams, and in mountains.
WHY: Scientists want to learn whether the glow helps salamanders find each other, pick a mate, or make their warning colors stand out to predators.

Why it matters

It shows you that even animals you might read about in a nature book can still hide surprises, so staying curious on a walk outside can lead to real discoveries.

Test what you remember

1. Where do fire salamanders live?

(A) In dry deserts
(B) Only in deep oceans
(C) In wet forests, near streams, and in mountains
(D) Only in city parks

2. What colors is the salamander's glow?

(A) Red and orange
(B) Green and cyan
(C) Black and yellow
(D) Purple and pink

3. Where on the body is the glow strongest?

(A) On the yellow underside of their bodies and along their sides
(B) On the tail tip only
(C) On their feet
(D) On the top of the head only

4. What does the glow come from?

(A) Their teeth
(B) Glands in their skin
(C) Food they eat
(D) Their eyes

5. How are fireflies different from fire salamanders in this story?

(A) Fireflies live only in mountains
(B) Fireflies glow only in water
(C) Fireflies have no color
(D) Fireflies make their own light

6. What do scientists not yet know about the salamanders?

(A) Where they live
(B) What color they are
(C) Why salamanders glow
(D) Whether they give off poison

Answer key (try the quiz first!): 1: C 2: B 3: A 4: B 5: D 6: C

Questions to think about

1. Bernat Burriel: It is fascinating that such a well-studied animal still holds secrets.

MY THOUGHTS (at least 20 words):